

DANIEL M. DINAKARAPANDIAN

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Research Interests: biomolecular assembly, amyloid oligomers, designer peptides, molecular dynamics, physical biology of disease.

EDUCATION

Ph.D., Chemical Engineering Aug 2022 – Present
Georgia Institute of Technology, Atlanta, GA. GPA: 3.73. Advisor: Prof. Anant K. Paravastu.

B.Tech., Chemical Engineering May 2018
Anna University, Chennai, India. GPA: 3.76. First Class with Distinction; Dean's List.

RESEARCH EXPERIENCE

Graduate Research Assistant, Paravastu Lab Aug 2022 – Present
School of Chemical & Biomolecular Engineering, Georgia Tech.

- *Thesis (in progress):* Controlling Alzheimer's Amyloid- β 42 Oligomer Formation by Pathway Selection
- Developed a protocol for preparing A β 42 oligomers in the quantity and quality needed for solid-state NMR (*Methods in Enzymology*, 2026).
- With the Wood and Raskatov labs, showed that side-chain deamidation at Q15 or N27 redirects A β 42 aggregation and dampens microglial inflammatory responses (*ACS Chemical Neuroscience*, 2025 and 2026).
- Apply solid-state NMR, cryo-EM, SEC, CD, FTIR, mass spectrometry (TIMS-TOF), and mass photometry to characterize A β 42 oligomers, fibrils, and designer α -helical assemblies.
- Run all-atom and coarse-grained MD simulations (NAMD, MARTINI 3, PRIME20/DMD) on A β (16–22) registry states and A β 42 self-assembly; develop Python tools for NMR data analysis and biophysical workflows.
- Mentor 10+ undergraduate researchers.

PUBLICATIONS

[†] Co-first author. Names in **bold** indicate the author of this CV.

Peer-Reviewed Articles & Book Chapters

1. Griffin, M.N.[†], **Dinakarapandian, D.M.**[†], Li, C., Ramaswamy, P., Saheba, S.J., Lee, J., Sudarshan, T.R., Sajimon, M., Wheeler, C.J., Raskatov, J.A., Paravastu, A.K., Wood, L.B. (2026). A β -42 side chain deamidation at Q15 or N27 modulates protein aggregation and alters microglial cytokines and CD68. *ACS Chemical Neuroscience*. DOI: [10.1021/acschemneuro.6c00024](https://doi.org/10.1021/acschemneuro.6c00024).
2. Dodd-o, J., Shadpoor, B., Roy, A., Jafari, R., Sudarshan, T.R., Holberton, A., He, D., Coppola, F., **Dinakarapandian, D.M.**, Malu, D., Gharati, G., Griffith, A., Rohe-Weiner, H., Patel, S., Bonder, E., Hu, Y., Wood, L., Paravastu, A.K., Kral, P., Kumar, V.A. (2026). Programming angiogenesis with tunable assemblies of multifunctional peptides. *Science Advances* **12**, eaef5064. DOI: [10.1126/sciadv.aef5064](https://doi.org/10.1126/sciadv.aef5064).
3. **Dinakarapandian, D.M.**, Watzlawik, J.O., Sudarshan, T.R., Rosenberry, T.L., Paravastu, A.K. (2026). How to prepare Alzheimer's amyloid- β (1–42) oligomer samples with sufficient quantity and quality for biophysical and solid-state NMR measurements. *Methods in Enzymology* **729**, 419–433. DOI: [10.1016/bs.mie.2026.02.002](https://doi.org/10.1016/bs.mie.2026.02.002).

4. Sajimon, M., Wheeler, C.J., Foley, A.R., Chan, K., Griffin, M.N., **Dinakarapandian, D.M.**, Holberton, A., Joy, J., Paravastu, A.K., Wood, L.B., Raskatov, J.A. (2025). Asparagine deamidation attenuates toxicity, aggregation, and microglial responses of Alzheimer's amyloid- β . *ACS Chemical Neuroscience* **16**, 4076–4085. DOI: [10.1021/acschemneuro.5c00544](https://doi.org/10.1021/acschemneuro.5c00544).
5. Mondal, M., Jankoski, P.E., Lee, L.D., **Dinakarapandian, D.M.**, Chiu, T.-Y., Swetman, W.S., Wu, H., Paravastu, A.K., Clemons, T.D., Rangachari, V. (2024). Reversible disulfide bond cross-links as tunable levers of phase separation in designer biomolecular condensates. *Journal of the American Chemical Society* **146**, 25299–25311. DOI: [10.1021/jacs.4c09557](https://doi.org/10.1021/jacs.4c09557).

Preprints

6. Das, A.[†], Wang, F.[†], **Dinakarapandian, D.M.**[†], Wang, S., Wong, K.M., Robang, A.S., Sudarshan, T.R., Li, J., Nijhawan, A.K., Zuo, X., Dunin-Horkawicz, S., Paravastu, A.K., Egelman, E.H., Conticello, V.P. (2026). Reprogramming coiled-coil design principles to create cross- α peptide filaments. *ChemRxiv*. DOI: [10.26434/chemrxiv.15006530/v1](https://doi.org/10.26434/chemrxiv.15006530/v1).

CONFERENCE PRESENTATIONS

Talks

- **Dinakarapandian, D.M.**, Ramaswamy, P., Griffin, M., Sudarshan, T.R., Wood, L., Paravastu, A.K. AIChE Annual Meeting 2025 (American Institute of Chemical Engineers). “Deamidation in Alzheimer's amyloid beta (1–42) redirects aggregation pathways and reduces toxic oligomer formation.”
- **Dinakarapandian, D.M.** Georgia Magnetic Resonance Symposium, hosted by Emory University, Atlanta, GA. “The distinction between amyloid fibril and oligomer assembly pathways for Alzheimer's amyloid- β peptides.” Dec 12, 2024. *Invited talk*.

Posters

- **Dinakarapandian, D.M.**, et al. ACS Spring 2026 (American Chemical Society), “Deamidation of amyloid beta(1–42) redirects aggregation pathway and alters microglial response.”
- **Dinakarapandian, D.M.**, Das, A., Robang, A.S., Conticello, V.P., Paravastu, A.K. SERMACS 2024 (Southeastern Regional Meeting of the American Chemical Society). “Combining solid-state NMR and cryo-EM to probe structure of a designer α -helical filament.” Presented at the 52nd Southeastern Magnetic Resonance Conference (SEMRC) session. **Best Poster Award**.

PEER REVIEW

Reviewer, <i>Communications Chemistry</i> (Springer Nature)	2026
Book chapter reviewer, <i>Fibrous Proteins: Structures and Mechanisms</i> (Springer Nature)	2025

AWARDS & HONORS

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| • Eckert Graduate Fellowship, School of Chemical & Biomolecular Engineering, Georgia Tech | 2026 |
| • Best Mentor Award, Nakatani Foundation/Georgia Tech | 2025 |
| • Best Poster Award, Southeastern Magnetic Resonance Conference (SEMRC) | 2024 |
| • Petit Mentorship Award, Georgia Tech | 2024 |
| • BASF Capstone Scholarship, Georgia Tech | 2024 |
| • First Class with Distinction, Department of Chemical Engineering, Anna University | 2018 |

INDUSTRY EXPERIENCE

Data Analyst, Alexa Shopping — Amazon Inc.
Chennai, India.

Oct 2021 – Aug 2022

- Conducted root-cause analysis of voice-query goal success/failure rates across US, UK, and AU domains; partnered with the Natural Language Understanding team to fix recurring failures in ASR, Intent, and Alexa neural-network components.
- Built 10+ automated workflows (GYRO module, Redshift) that streamlined data-processing pipelines for ASR diagnostics.

Process Engineer — Manali Petrochemicals Ltd.
Chennai, India.

Oct 2018 – Sep 2021

- Drove plant-productivity projects across adsorption towers, multi-effect evaporators, and distillation columns (pressurized and vacuum) to improve capacity and reduce cost.
- Improved plant data-analytics integration with Yokogawa Centum-VP DCS, reducing downtime and improving cost efficiency.

LEADERSHIP & SERVICE

Nakatani Graduate Mentor

Fall 2025 & Fall 2026

Nakatani Research & International Experiences for students, Nakatani Foundation/Georgia Tech.

Professional Development Chair, IChE Student Chapter

Jan – Jul 2018

Indian Institute of Chemical Engineers, Anna University.

TECHNICAL SKILLS

Structural & Biophysical: solid-state NMR; solution NMR; cryo-EM; SEC/FPLC; RP-HPLC; circular dichroism; FTIR; LC-MS; native mass spectrometry (TIMS-TOF); mass photometry; SDS-PAGE; solid-phase peptide synthesis (Fmoc, CEM Liberty Blue).

Computational: Python, R, MATLAB, Mathematica; NAMD, VMD, GROMACS; MARTINI 3, CALVADOS 2; SPINEVOLUTION; Packmol, MDAnalysis, DSSP; Aspen.